Nanodisc models for calculation of small angle scattering patterns

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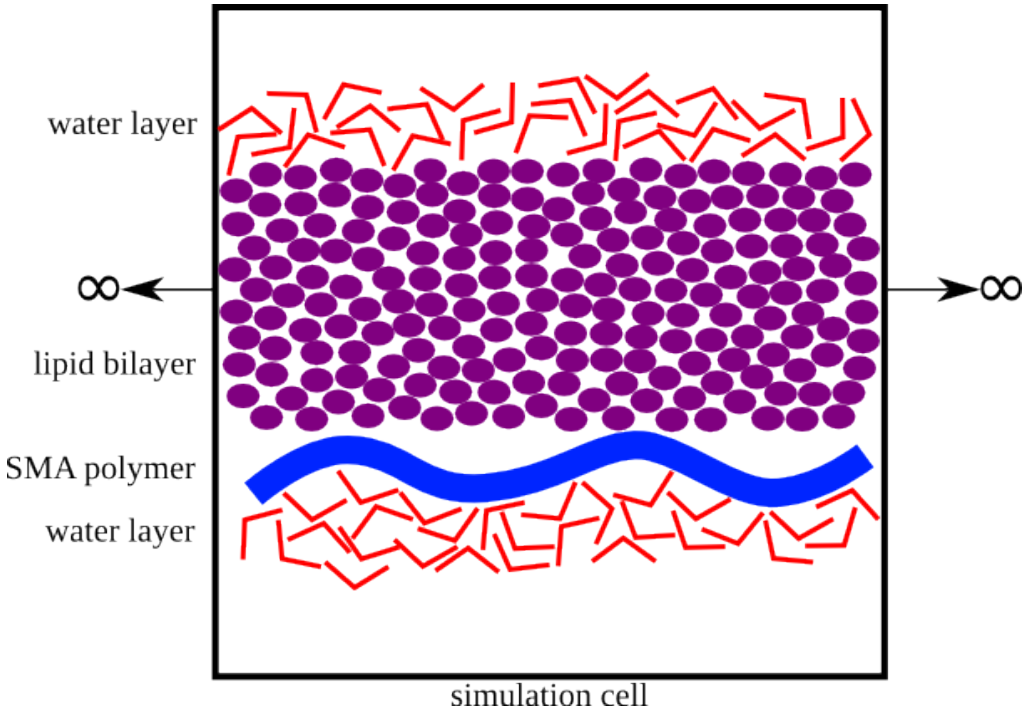
Introduction

- The ability to efficiently model a protein-lipid-SMA complex would allow for better data analysis
- The neutron/X-ray scattering pattern of such a system can then be found using an online utility, such as SASSIE-web
- Ryan Franklin and Joseph Curtis started this work, and I will be continuing it over in the coming months

Infinite lipid rod

- A small block of DMPC was positioned to cross through the periodic boundary of the cell
- This allowed for the creation of an *effectively infinite* bilayer
- Allowing for a short simulation time to study the interaction, while keeping the properties of the similar to that of the disc



The simulation cell for the *infinite* lipid rod

Pure styrene polymer interaction

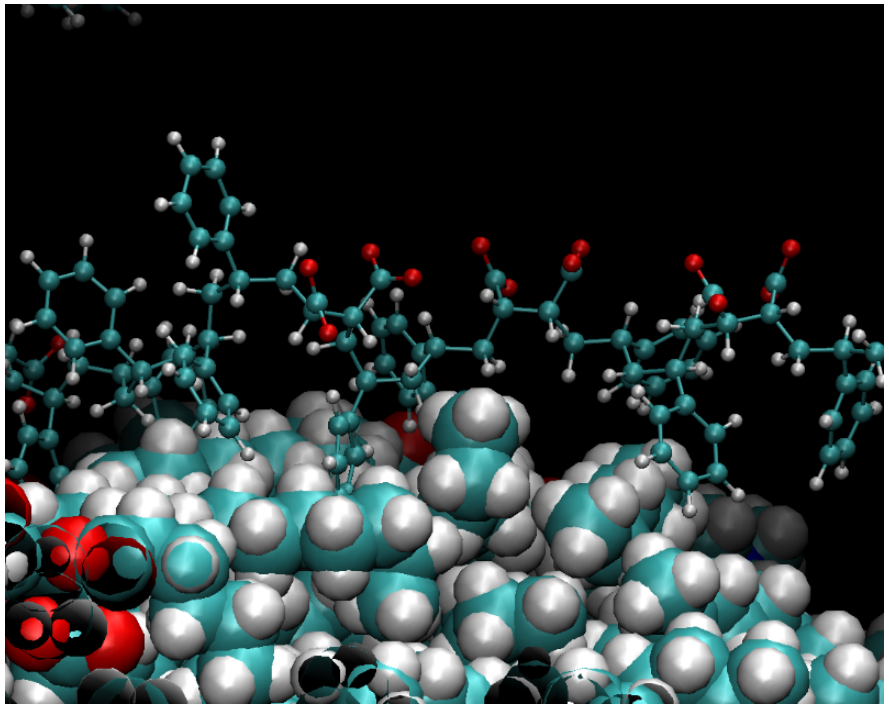
- A polymer of pure styrene was constructed to test that the interaction between the phenyl ring and the lipid tails
- The phenyl rings intercalate quickly into the lipid tails

H.264 mp4

A 200 ps NPT simulation

Random composition SMA polymer

- The interaction between the random polymer and this *infinite* lipid bilayer was also investigated
- Many of the phenyl groups of the styrene subunits intercalated with the lipid tails
- As expected, the maleic acid subunits were orientated towards the water layer



A 200 ps NPT simulation

Building wrapped nanodiscs

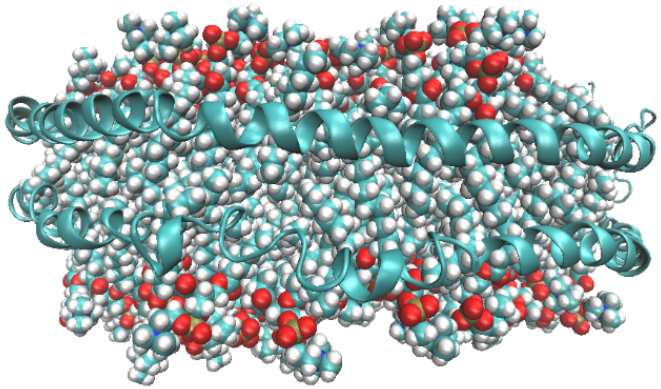
- Following success the *infinite* lipid bilayer, the process was 'scaled-up' to a full nanodisc
- Without the polymer/protein wrapping, it was possible to reproduce the packing density and dimensions of a lipid nanodisc *in-silico*
- However, this was a non-trival task involving many stages and relaxations

H.264 mp4

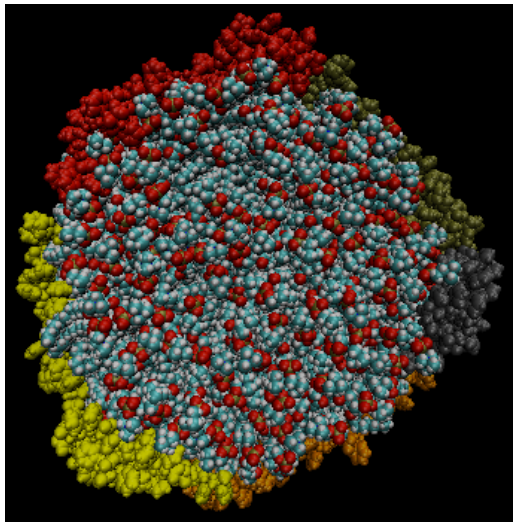
The lipid nanodisc

Building wrapped nanodiscs

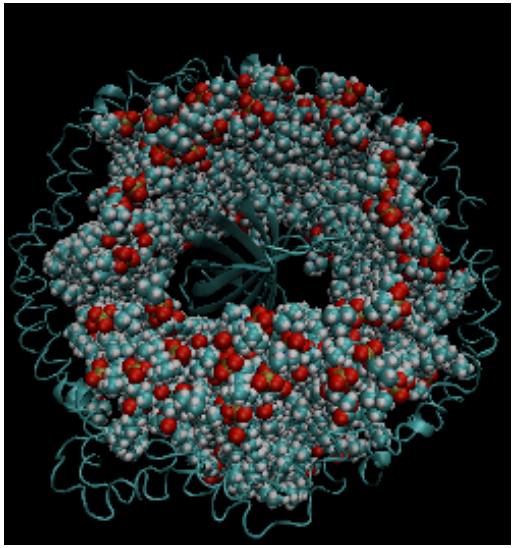
- Currently Ryan & Joseph have managed to wrap one of their nanodiscs in MSP and are currently finalising work on wrapping with the SMA polymer
  - They were able to successfully generate a small angle scattering pattern from the MSP wrapped nanodisc
- They have also manages to build a lipid nanodisc with the HIV-gag protein in it, wrapped with MSP.



MSP



SMA



MSP w/ HIV-gag

Future Work

- I will be working with Joseph Curtis to use his SASSIE package to generate small angle scattering patterns for these nanodiscs
- We will use this procedure to generate nanodiscs containing a wide variety of proteins, in particular those studied by Cecilia Tognoloni
- I hope to use my coarse-graining expertise to reduce the computational effort required to generate these structures, increasing the accessability of such modelling
  - Joseph has started this work, applying MARTINI coarse graining to the protein wrapped nanodisc

Thanks for listening

I am happy to answer any questions

My family dog, Sadie